Soft Robotic Grippers using Tactile Sensor

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1. Introduction

Robotic manipulation systems are traditionally designed around rigid mechanical structures that provide predictable motion, positioning accuracy, and controlled grasping forces. Such grippers perform effectively when the objects being handled have well-defined geometries, known positions, and relatively consistent mechanical properties. However, these assumptions become difficult to maintain when the objects are fragile, irregularly shaped, or highly variable. Excessive contact force or localized stress can damage the object, while insufficient force can result in unstable grasping and object slip. These limitations have increased the need for compliant robotic end-effectors that can adapt their shape to the object during interaction rather than relying entirely on precise rigid-body positioning. The project report considers applications including automated agriculture, medical logistics, and explosive ordnance disposal (EOD), where safe interaction with delicate or uncertain objects is an important requirement.

Soft robotic grippers provide a different approach to manipulation by replacing conventional rigid linkages with compliant structures capable of undergoing relatively large deformations. This compliance allows the gripper to conform passively to the surface of an object and distribute contact forces over a larger area. The proposed system uses a 3D-printed compliant structure fabricated using stereolithography (SLA) or digital light processing (DLP), allowing complex geometries and internal features to be incorporated into the gripper. The report considers both pneumatic actuation and embedded Shape Memory Alloy (SMA) actuators as possible methods for producing finger deformation. While this mechanical compliance improves adaptability, it also introduces nonlinear material behaviour, deformation-dependent response, hysteresis, and fatigue-related limitations that are less prominent in conventional rigid grippers.

An additional challenge is obtaining reliable information about the contact between the gripper and the manipulated object. Conventional robotic systems commonly depend on external cameras for object localization and grasp evaluation. Such systems can become less effective when visibility is poor or when rapid manipulation is required. The proposed design therefore incorporates tactile sensing directly into the gripper using thin Force Sensitive Resistors (FSRs), with the FSR402 identified as the intended sensor type. The sensors are positioned at the grasping surfaces so that changes in contact force can be measured locally. This provides information about the interaction between the gripper and the object without requiring the entire grasping process to be inferred from an external camera.

The use of compliant photopolymer structures, however, makes both mechanical design and control more demanding. Unlike a rigid component whose deformation can generally be described using linear elastic assumptions over a limited operating range, soft elastomeric structures can undergo large and nonlinear deformation. Consequently, conventional linear material models are not sufficient for accurately representing the behaviour of the proposed

gripper. The report identifies hyperelastic constitutive models, including Neo-Hookean, Mooney-Rivlin, Yeoh, and Ogden formulations, as relevant approaches for describing the nonlinear response of the printed material. Finite Element Analysis (FEA) using Ansys Mechanical is therefore required to study deformation and stress distribution and to support the design of the compliant structure. Mesh selection, large deformation behaviour, volumetric locking, and stress concentration around highly deformed regions are among the computational issues that must be considered during this process.

Material durability presents another limitation. The layer-by-layer fabrication of 3D-printed elastomers can introduce microscopic defects and imperfections that may become stress concentration sites under repeated loading. Since a robotic gripper is expected to undergo repeated grasping and releasing operations, cyclic loading can gradually affect the integrity of the compliant structure. The design must therefore consider stress concentrations and appropriate geometric features in regions subjected to repeated deformation. Post-processing is also important, particularly for designs containing internal cavities or channels, where residual uncured resin can affect the long-term behaviour of the printed component.

The actuation mechanism introduces a separate set of control challenges. Pneumatic actuation can produce bending through controlled pressurization of internal chambers, while SMA actuation can generate deformation through the thermomechanical response of embedded Nitinol wires. In the latter case, the electrical current supplied to the SMA must be controlled carefully because its response is strongly dependent on temperature and thermal history. In addition, both the SMA actuator and the compliant polymer structure exhibit hysteresis. As a result, the same input does not necessarily produce the same output during loading and unloading. If this behaviour is not considered, the accuracy of position or force control can be reduced. The report therefore proposes the use of a Prandtl-Ishlinskii hysteresis model and its inverse for compensating the nonlinear response before applying the control command to the actuator.

Tactile measurements introduce further signal-processing requirements. The raw output of the FSRs can contain mechanical vibrations, electrical interference, and other parasitic components that can obscure useful contact information. A signal-conditioning stage is consequently required before the data are processed by the embedded controller. The proposed architecture uses an analog interface followed by ADC acquisition and digital filtering. A second-order Butterworth low-pass filter is specified for reducing unwanted high-frequency components while retaining the relevant force information. For slip detection, however, high-frequency information that would normally be removed from a force measurement may itself be useful. The proposed system therefore considers frequency-domain analysis of the tactile signal using a Fast Fourier Transform (FFT) to identify vibration characteristics associated with incipient slip.

The central gap addressed by the project is therefore not limited to the mechanical design of a soft gripper. A compliant structure alone does not provide sufficient information or control capability for reliable manipulation. The mechanical deformation, tactile sensing, actuator nonlinearities, signal processing, and embedded communication must operate as an integrated system. The project proposes an architecture in which tactile information is processed close to the point of interaction rather than relying entirely on computationally intensive external visual feedback. This requires an embedded platform capable of handling sensor acquisition, signal processing, control, and communication within predictable timing constraints. The proposed implementation uses an ESP32 with FreeRTOS for task management and micro-ROS for communication with the broader ROS 2 environment.

The present work consequently develops a framework for the design and implementation of an advanced 3D-printed compliant soft robotic gripper with embedded tactile sensing. The mechanical design is considered through CAD and nonlinear FEA in Ansys Mechanical, with hyperelastic material models used to represent the compliant structure. The gripper is intended to incorporate either pneumatic actuation or SMA-based actuation, together with FSR-based tactile sensing. On the embedded side, the ESP32-based architecture uses FreeRTOS for concurrent task execution, ADC-based sensor acquisition, digital filtering, and FFT-based signal analysis. The control framework incorporates hysteresis modeling and inverse compensation using the Prandtl-Ishlinskii formulation, while micro-ROS provides communication between the embedded controller and the ROS 2 system.

The main objective of the project is to establish an integrated soft robotic gripper architecture capable of compliant object manipulation while obtaining tactile information directly from the grasping interface. The scope covers the mechanical and computational design of the compliant gripper, nonlinear FEA of its structure, integration of actuation and FSR-based sensing, embedded signal processing, hysteresis compensation, and real-time communication. Particular emphasis is placed on addressing the nonlinear mechanical behaviour and sensing limitations associated with soft robotic systems. The resulting framework provides the basis for evaluating how compliant mechanical design, tactile feedback, nonlinear modeling, and deterministic embedded control can be combined within a single robotic end-effector.